

OBSERVING THE SKIES

LIGHT IS OUR MAIN SOURCE OF INFORMATION ABOUT DISTANT OBJECTS IN THE UNIVERSE, AND GROUND-BASED TELESCOPES REMAIN IMPORTANT TOOLS FOR CAPTURING LIGHT AND STUDYING FAR-AWAY STARS AND GALAXIES.

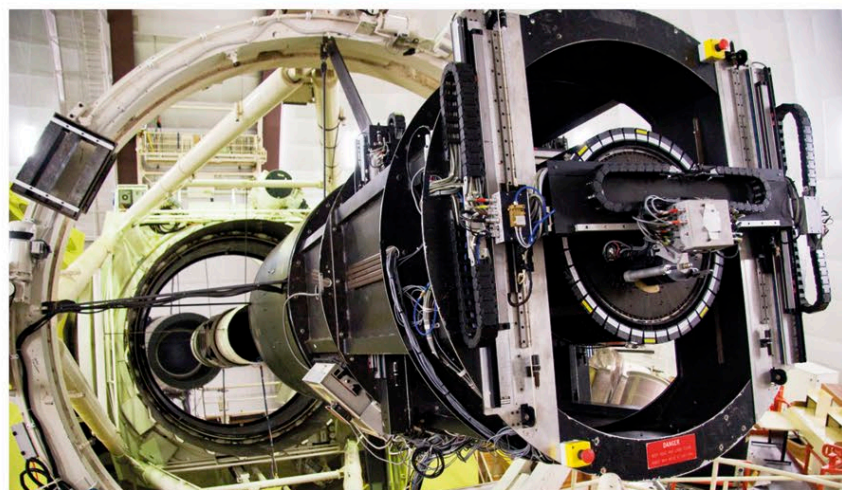
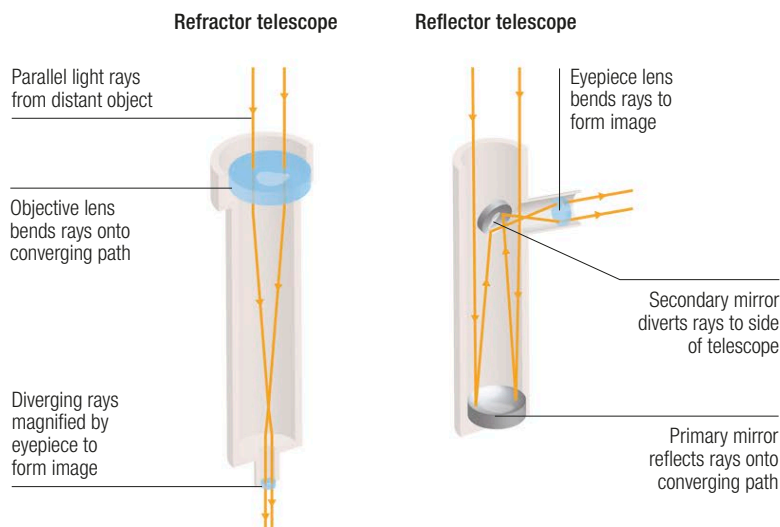
Very few objects from space ever reach Earth, and most of those only come from our immediate planetary neighborhood. Studying light that reaches our planet from distant space is therefore one of the best ways of learning about objects in the wider Universe. Many other forms of radiation are absorbed by Earth's atmosphere and astronomers use space-based observatories (see pp.80–81) to study objects in these other wavelengths.

By gathering light across a large surface and concentrating it into a much smaller image, telescopes allow us to see objects that are much fainter than those we can see with the

unaided eye. Magnifying these small images then enables us to distinguish much finer detail. However, modern scientific telescopes are very different machines from those used by backyard stargazers. Most are reflector designs that use a series of mirrors to bring light onto converging paths and create a focused image on a detector instrument. The telescope is supported by a mount or cradle that swings back and forth and allows it to keep pace with the path of objects across the sky. A technique called interferometry allows astronomers to link two or more telescopes together and detect even finer details.

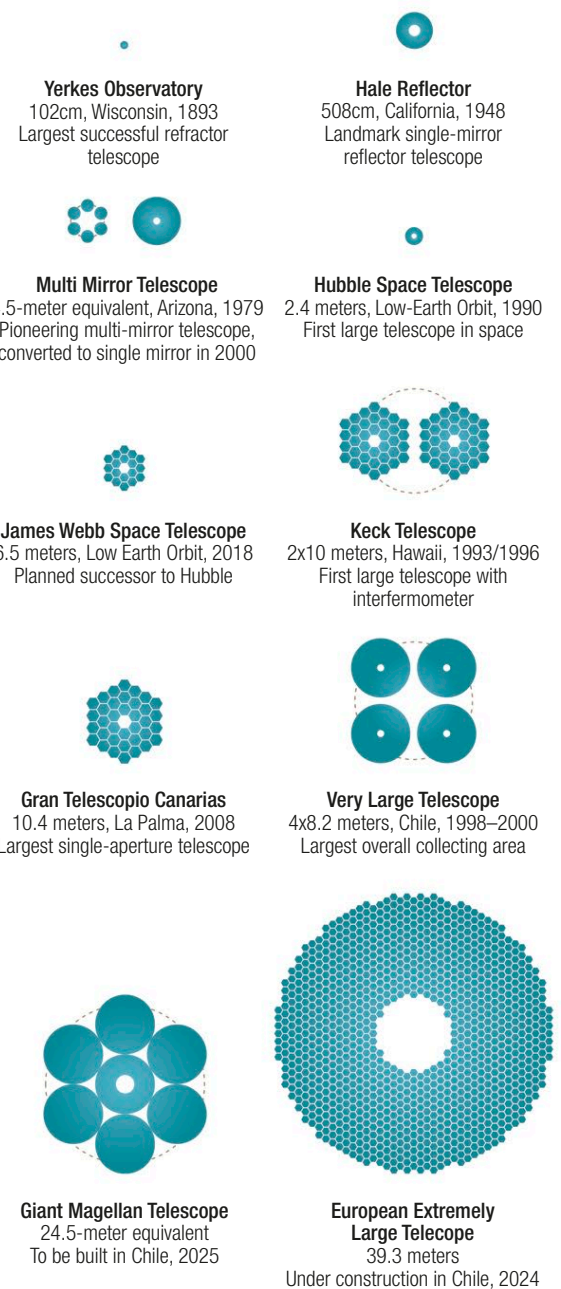
▷ Refractor and reflector telescopes

The first telescopes, invented by Dutch eyeglass-makers in the Netherlands around 1609, used a lens-based refractor design. One lens (the objective) collects light and bends it to a focus, while another (the eyepiece) magnifies image. The simplest reflector designs, invented by Isaac Newton around 1668, use a curved mirror to collect light and direct it to a secondary mirror, which then directs it to a lens-based eyepiece.



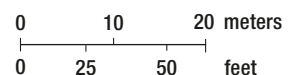
◁ Learning from light

Professional astronomers rarely use an eyepiece to look directly through a telescope. Instead, they use the instrument to channel light to various detectors. These can include digital cameras, photometers (which precisely measure the brightness of light from individual objects), and spectrometers that analyse colors of light, enabling scientists to learn about the chemistry of stars.



△ Collecting areas

Telescope collecting areas were limited by technology for much of the 20th century, but instruments have grown rapidly in the past few decades.



A typical 8-in (20-cm) reflecting telescope gathers **830 times more light** than the human eye alone